



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

Experiment -8

Student Name: Harsh

Branch: BE-CSE

Semester: 6th

Subject Name: Project-Based Learning in
Java with Lab

UID: 22BCS12488

Section/Group: IOT_NTPP_603-B

Date of Performance: 17/03/2025

Subject Code: 22CSH-359

7.1.1.Aim: To develop a servlet that accepts user credentials from an HTML form and displays a personalized welcome message on successful login.

7.1.2 Objective: Learn form handling with Servlets
Understand HTTP request/response handling
Practice doPost() method

7.1.3 Code:

```
<!DOCTYPE html>
<html>
<head><title>Login</title></head>
<body>
  <form action="LoginServlet" method="post">
    Username: <input type="text" name="username"><br>
    Password: <input type="password" name="password"><br>
    <input type="submit" value="Login">
  </form>
</body>
</html>
```

```
import java.io.*; import
javax.servlet.*;
import javax.servlet.http.*;
```

```
public class LoginServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {        String user =
    request.getParameter("username");
        String pass = request.getParameter("password");
```

```
response.setContentType("text/html");
PrintWriter out = response.getWriter();

if ("admin".equals(user) && "1234".equals(pass)) {
    out.println("<h2>Welcome, " + user + "!</h2>");
} else {
    out.println("<h2>Login Failed. Invalid credentials.</h2>");
}
}
}

<web-app>
<servlet>
    <servlet-name>LoginServlet</servlet-name>
    <servlet-class>LoginServlet</servlet-class>
</servlet>
<servlet-mapping>
    <servlet-name>LoginServlet</servlet-name>
    <url-pattern>/LoginServlet</url-pattern>
</servlet-mapping>
</web-app>
```

Output:

- 1) On correct login: Welcome, Sarthak !
- 2) On failure: Login Failed. Invalid credentials.

7.2.1 Aim: To build a servlet integrated with JDBC that displays all employees and enables search by employee ID.

Objective: 1) Use JDBC with Servlet 2)

Fetch and display records

- 3) Implement search functionality

7.2.2 Code:

```
<!DOCTYPE html>
```

```
<html>
<head><title>Search Employee</title></head>
<body>
  <form action="EmployeeServlet" method="post">
    Enter Employee ID: <input type="text" name="empId">
    <input type="submit" value="Search">
  </form>
</body>
</html>
```

```
import java.io.*; import
javax.servlet.*; import
javax.servlet.http.*; import
java.sql.*;
```

```
public class EmployeeServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
```

```
        String empId = request.getParameter("empId");
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
```

```
        try {
            Class.forName("com.mysql.jdbc.Driver");
            Connection con =
            DriverManager.getConnection("jdbc:mysql://localhost:3306/company", "root", "password");
```

```
            String query = "SELECT * FROM employees WHERE emp_id=?";
            PreparedStatement ps = con.prepareStatement(query);
            ps.setString(1, empId);
            ResultSet rs = ps.executeQuery();
```

```
            if (rs.next()) {
                out.println("<h2>Employee Details</h2>");
                out.println("ID: " + rs.getInt(1) + "<br>");
                out.println("Name: " + rs.getString(2) + "<br>");
                out.println("Department: " + rs.getString(3));
```

```
        } else {  
            out.println("No employee found with ID " + empId);  
        }  
  
        con.close();    }  
catch (Exception e) {  
    out.println("Error: " + e.getMessage());  
}  
}  
}
```

7.2.3 Output:

1) Enter an employee ID → Shows details if found.

2) Not found → "No employee found with ID X"

7.3.1 Aim: To develop a JSP-based student portal that accepts attendance data and saves it to the database using a servlet.

Objective: 1) Combine JSP for UI and Servlets for logic

2) Perform INSERT using JDBC

3) Build a real-world web flow

Code:

```
<%@ page language="java" %>  
<html>  
<head><title>Student Attendance</title></head>  
<body>  
    <h2>Mark Attendance</h2>  
    <form action="AttendanceServlet" method="post">  
        Roll No: <input type="text" name="roll"><br>  
        Name: <input type="text" name="name"><br>  
        Status: <select name="status">  
            <option>Present</option>
```

```
<option>Absent</option>
</select><br>
<input type="submit" value="Submit">
</form>
</body>
</html>
```

```
import java.io.*; import
javax.servlet.*; import
javax.servlet.http.*; import
java.sql.*;
```

```
public class AttendanceServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {        String roll =
request.getParameter("roll");
        String name = request.getParameter("name");
        String status = request.getParameter("status");

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
```

```
        try {
            Class.forName("com.mysql.jdbc.Driver");
            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/student_portal", "root",
"password");

            String query = "INSERT INTO attendance (roll_no, name, status) VALUES (?, ?,
?)";
            PreparedStatement ps = con.prepareStatement(query);
            ps.setString(1, roll);        ps.setString(2, name);
            ps.setString(3, status);

            int i = ps.executeUpdate();
            if (i > 0) {
                out.println("<h3>Attendance marked successfully for " + name + "!</h3>");
            }
        }
```

```
        con.close();  
    } catch (Exception e) {  
        out.println("Error: " + e.getMessage());  
    }  
}  
}
```

```
CREATE TABLE attendance (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    roll_no VARCHAR(20),  
    name VARCHAR(100),  
    status VARCHAR(10)  
);
```

OUTPUT

Form submission → "Attendance marked successfully for John!" And the data is stored in the database.